

Time : 3 Hours]

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

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Q1. Objective type questions. (Compulsory Question)

1. If $L\{f(t)\} = f(s)$ then $L\{f'(t)\}$ is equal to.....

- a) $sL\{f(t)\}$
- b) $sL\{f(t)\} - f(0)$
- c) $sL\{f(t)\} - f'(0)$
- d) none of these

2. The $L\{e^{at} - \cos bt\} = \dots\dots\dots$

- a) $\frac{1}{s-a} - \frac{s}{s^2+b^2}$
- b) $\frac{1}{s-a} + \frac{s}{s^2+b^2}$
- c) $\frac{1}{s+a} - \frac{s}{s^2+b^2}$
- d) $\frac{1}{s-a} - \frac{s}{s^2-b^2}$

3. If $f(t)$ is Periodic function of Period T then $L\{f(t)\}$ is equal to.....

- a) $\frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$
- b) $\frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$
- c) $\frac{1}{1-e^{-sT}} \int_0^\infty e^{-st} f(t) dt$
- d) $\frac{1}{1-e^{-sT}} \int_0^\infty e^{-st} f(t) dt$

4. The inverse Laplace transform of $\frac{1}{s^2}$ is

- a) t^2
- b) 1
- c) t
- d) none of these

5. If $L^{-1}\{\bar{f}(s)\} = f(t)$ then $L^{-1}\left\{\frac{d}{ds} \bar{f}(s)\right\} = \dots\dots\dots$

- a) 0
- b) $f(t)$
- c) $t f(t)$
- d) $-t f(t)$

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If $L^{-1}\{f(s)\} = f(t)$ and $L^{-1}\{g(s)\} = g(t)$ then $L^{-1}\{f(s)g(s)\} = \dots$

- a) $\int_0^t f(u)g(t-u)du$ b) $\int_0^\infty f(u)g(t-u)du$
 c) $\int_0^t f(t)g(t)dt$ d) none of these

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If $F\{f(x)\} = F(s)$ then $F\{f(x-a)\}$ is equal to.....

- a) $e^{isa}F(s)$ b) $e^{-isa}F(s)$ c) $e^{isx}F(s)$ d) $e^{-isx}F(s)$

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The Fourier transform of $f(x)$ is $F(s)$ then the inversion formula is.....

- a) $f(x) = \frac{1}{\pi} \int_{-\infty}^{\infty} F(s)e^{-isx}ds$ b) $f(x) = \frac{1}{2\pi} \int_0^\infty F(s)e^{-isx}ds$
 c) $f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(s)e^{-isx}ds$ d) none of these

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The partial differential equation obtained by eliminating the arbitrary constants from $z = ax + by + ab$ is.....

- a) $xyz = pq$ b) $z = px - qy - pq$
 c) $z = px + qy + pq$ d) none of these

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The solution of partial differential equation $xp + yq = z$ is.....

- a) $f(x, y) = 0$ b) $f(x^2, y^2) = 0$
 c) $f(xy, yz) = 0$ d) $f\left(\frac{x}{y}, \frac{y}{z}\right) = 0$

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The Cauchy-Riemann equations for $f(z) = u(x, y) + iv(x, y)$ to be analytic are

- a) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$; $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$ b) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$; $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$
 c) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$; $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$ d) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial x}$; $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial y}$

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The singularities of $f(z) = \frac{\cos \pi z}{(z-1)(z-2)^2}$ is.....

- a) $z = -1, -2$ b) $z = 1, 2$ c) $z = 0, 1$ d) $z = 1, \pi$

Q2. Solve the following.

A) Find $L\left\{\int_0^t \frac{e^u \sin u}{u} du\right\}$ 6

B) Using Laplace transformation, evaluate $\int_0^\infty e^{-2t} t \sin 3t dt$ 6

Q3. Solve the following.

A) Using partial fraction method, find the inverse Laplace transform of $\frac{4s+5}{(s+2)(s-1)^2}$ 6

B) Using Laplace transform, solve $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 3y = e^{-t}$, given $y(0) = 1$ and $y'(0) = 0$. 6

Q4. Solve Any Two of the following.

A) Express $f(x) = \begin{cases} 1; & 0 \leq x \leq \pi \\ 0; & x > \pi \end{cases}$ as a Fourier sine integral and hence evaluate $\int_0^\infty \left(\frac{1-\cos\pi\lambda}{\lambda}\right) \sin\lambda\pi \, d\lambda$. 6

B) Find the Fourier sine and cosine Transform of $f(x) = e^{-ax}$, $a > 0$. 6

C) Using Parseval's identity, Show that, $\int_0^\infty \frac{dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{2ab(a+b)}$. 6

Q5. Solve Any Two of the following.

A) Find the general solution of the following differential equation. $p + 3q = 5z + \tan(y - 3x)$. 6

B) Form the partial differential equation by eliminating the arbitrary functions f and g from $y = f(x - at) + g(x + at)$. 6

C) Determine the solution of one-dimensional heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ where the boundary conditions are $u(x, 0) = 3\sin n\pi x$, $u(0, t) = 0$, $u(l, t) = 0$ where $0 < x < l$. <https://www.batuonline.com> 6

Q6. Solve Any Two of the following.

A) Show that $u = x^2 - y^2 - 2xy - 2x + 3y$ is a harmonic function and hence determine the analytic function, $f(z)$ in terms of z . 6

B) Using Cauchy's integral formula, evaluate $\oint_C \frac{\cos\pi z^2}{(z-1)(z-2)} dz$, where C is the unit circle $|z| = 3$. 6

- C) Find the residues of $f(z) = \frac{2z-1}{z(z+1)(z-3)}$ at its poles
inside the circle $|z| = 3$

*** End ***